

Documentation for Simple Snowmobile Physics

Namespace: Snowmobile

Script 1: SnowmobileExampleInputs

This script handles the input for the snowmobile, setting throttle and steering based on user input.

Variables:

- private SnowmobilPhysics snowmobile;; Reference to the SnowmobilPhysics component.

Methods:

- void Start(): Gets the reference to the SnowmobilPhysics component attached to the same GameObject.
- void Update(): Sets the throttle and steering inputs using the vertical and horizontal axes from user input.

Script 2: SnowmobilPhysics

This script manages the physics and movement of the snowmobile, including its motor, steering, and audio.

Serialized Fields:

- float maxSteering = 20f: Maximum steering angle.
- float maxVelocity = 50f: Maximum velocity.
- Vector3 centerOfMass: Center of mass for the snowmobile.
- float motorSmoothing = 1f: Smoothing factor for motor input.
- float volumeOffset = 0.25f: Base volume offset for motor audio.

- float volumeFactor = 1f: Factor by which volume increases with motor value.
- float pitchOffset = 1f: Base pitch offset for motor audio.
- float pitchFactor = 5f: Factor by which pitch increases with motor value.
- float wheelToTrackFactor = 0.075f: Factor for track animation based on wheel movement.
- Rigidbody wheelRB: Rigidbody of the wheel.
- ConfigurableJoint wheel: Configurable joint for the wheel.
- ConfigurableJoint leftSkii: Configurable joint for the left ski.
- ConfigurableJoint rightSkii: Configurable joint for the right ski.
- Transform steering: Transform for the steering object.

Private Variables:

- float motorValue = 0f: Current motor value.
- AudioSource source: AudioSource component for motor sound.
- TrackAnimation trackAnimation: Reference to the TrackAnimation component.
- Rigidbody rb: Rigidbody component of the snowmobile.
- float throttleInput: Current throttle input.
- float steeringInput: Current steering input.

Methods:

- void Start():
 - Disables inter-collision between all colliders in child objects.
 - Gets references to TrackAnimation, AudioSource, and Rigidbody components.

- **void Update():**
 - Sets the center of mass for the Rigidbody.
 - Adjusts the target rotation for the skis based on steering input.
 - Smooths the motor value and clamps it.
 - Sets the target angular velocity for the wheel.
 - Updates track animation based on wheel angular velocity.
 - Synchronizes steering animation.
 - Adjusts the volume and pitch of the motor audio based on motor value.
- **public void SetInputs(float throttle, float steering):** Sets the throttle and steering input values.

Usage

1. Attach both SnowmobileExampleInputs and SnowmobilPhysics scripts to the snowmobile GameObject.
2. Configure the serialized fields in the Inspector for proper tuning.
3. Run the scene and control the snowmobile using the vertical and horizontal inputs.

This setup allows for a basic implementation of snowmobile control, including input handling, physics management, and audio feedback.

